21CYB101J

Theory of coordination complexes – Bonding in coordination complexes

AB + CD

→ CoCl3.6NH3

CoCl3.5NH3

CoCl3.4NH3

CoCl3.3NH3

→

→ Experimental observations

↓

Postulates

↓

Theory

WERNER's THEORY

→ Limitations → ? → New theories

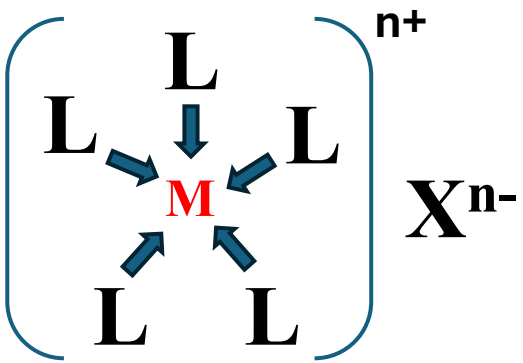
↓ Valence bond theory

↓ Crystal field theory

↓ Molecular orbital theory

Crystal field theory

Crystal field theory (CFT) is a bonding model that explains many important properties of transition-metal complexes, including their **colors, magnetism, structures, stability, and reactivity**.

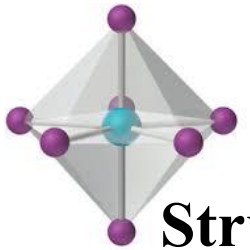


Color

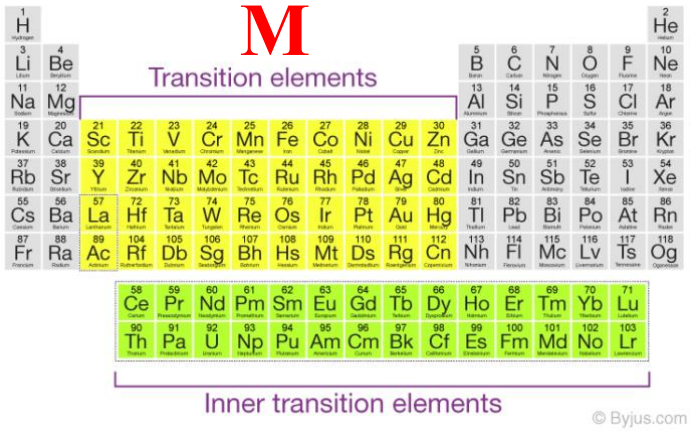
Magnetism

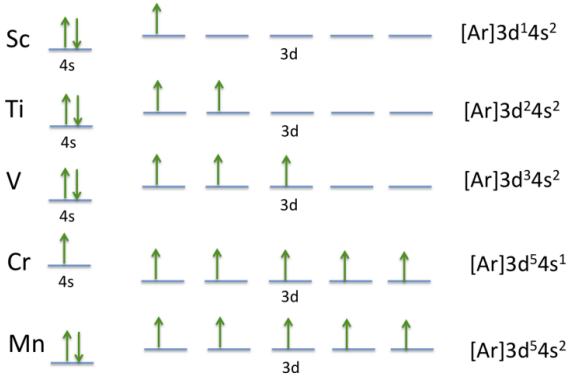
CFT

Stability and reactivity

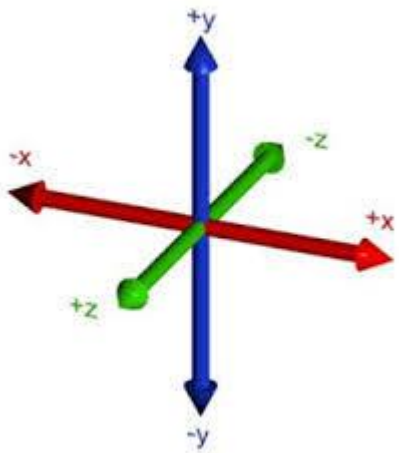


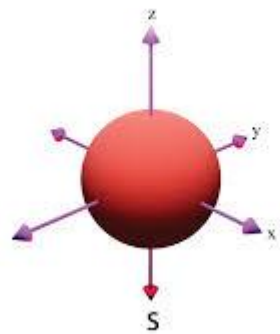
Structure

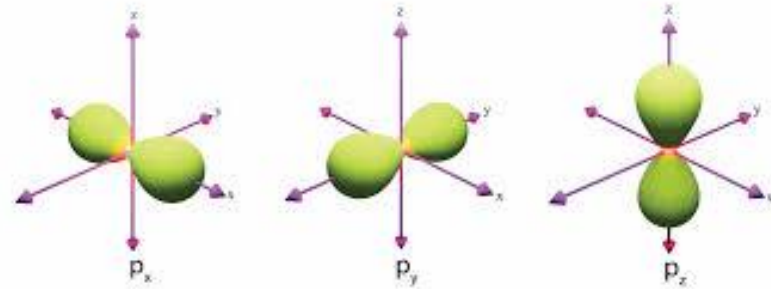


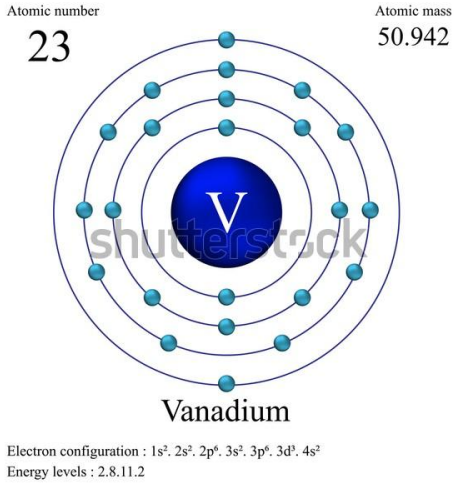


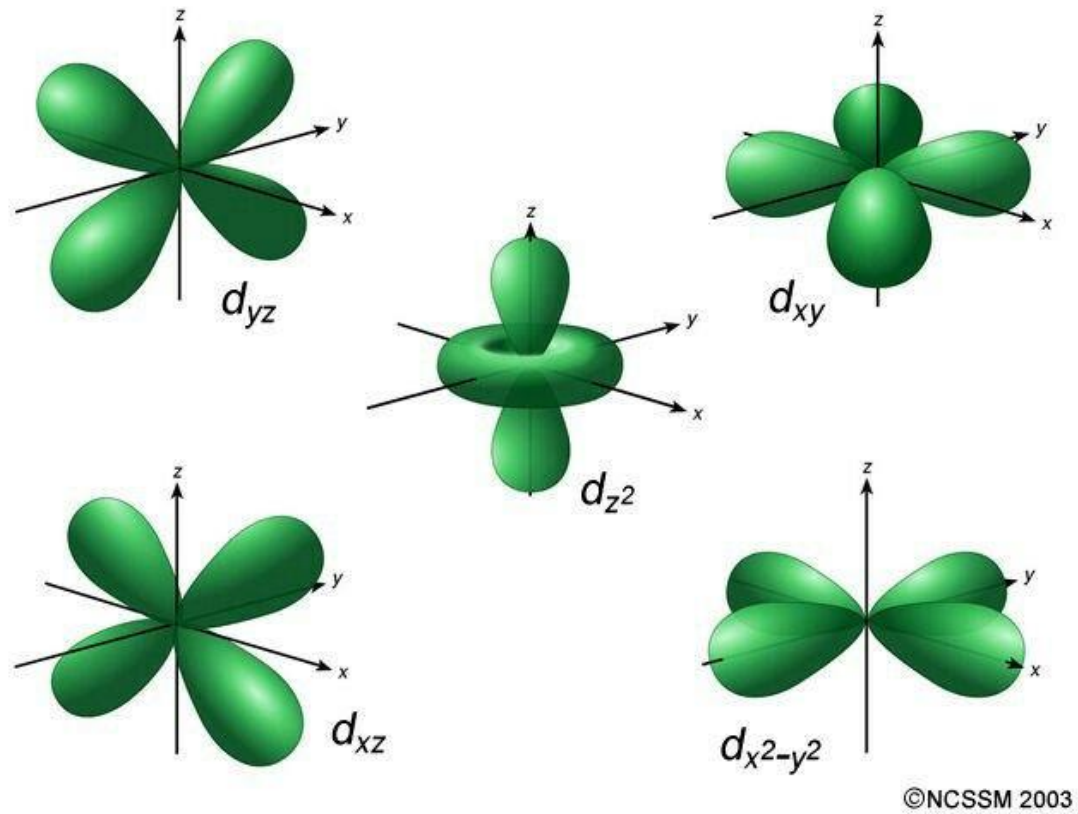
ORBITALS


+y
-x
-z
+x
+z
-y

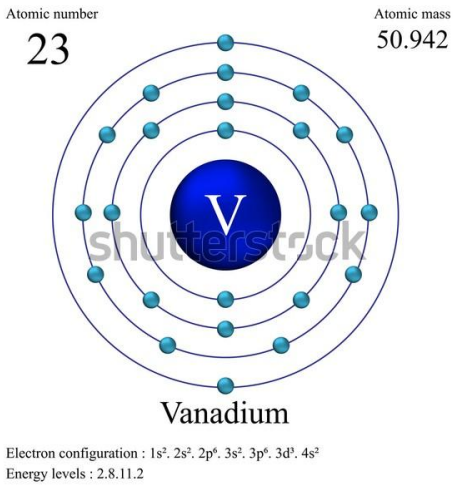

z
y
x
s

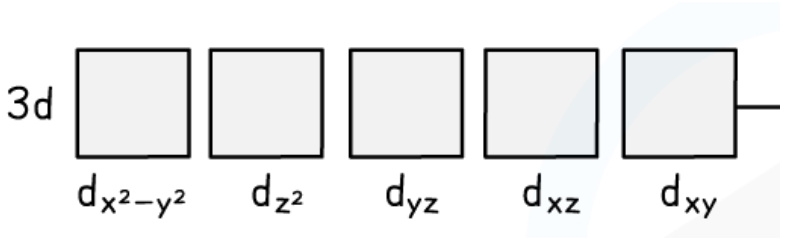

p_x
p_y
p_z


Atomic number
23
Atomic mass
50.942
V
shutterstock
Vanadium
Electron configuration : 1s². 2s². 2p⁶. 3s². 3p⁶. 3d³. 4s²
Energy levels : 2.8.11.2

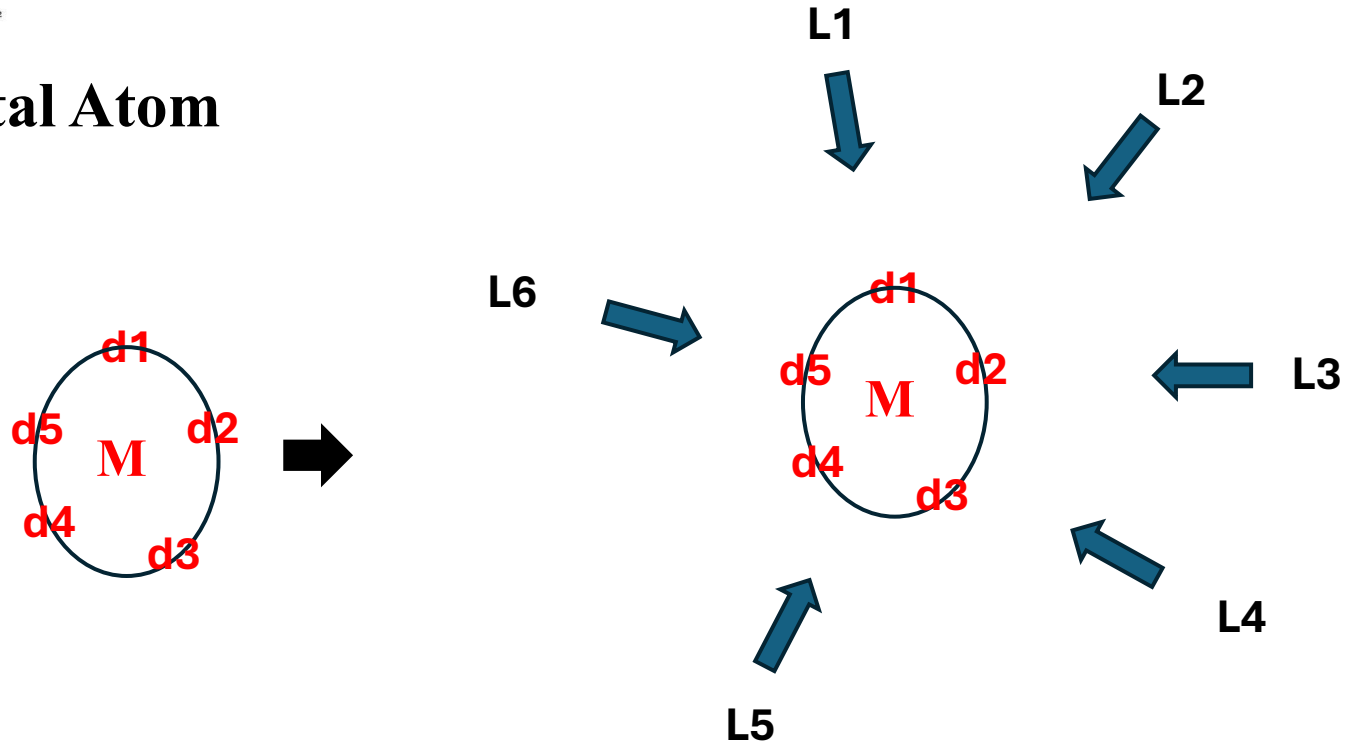

d_{yz}
d_{xy}
d_{z^2}
d_{xz}
$d_{x^2-y^2}$
©NCSSM 2003

Degenerate d orbitals





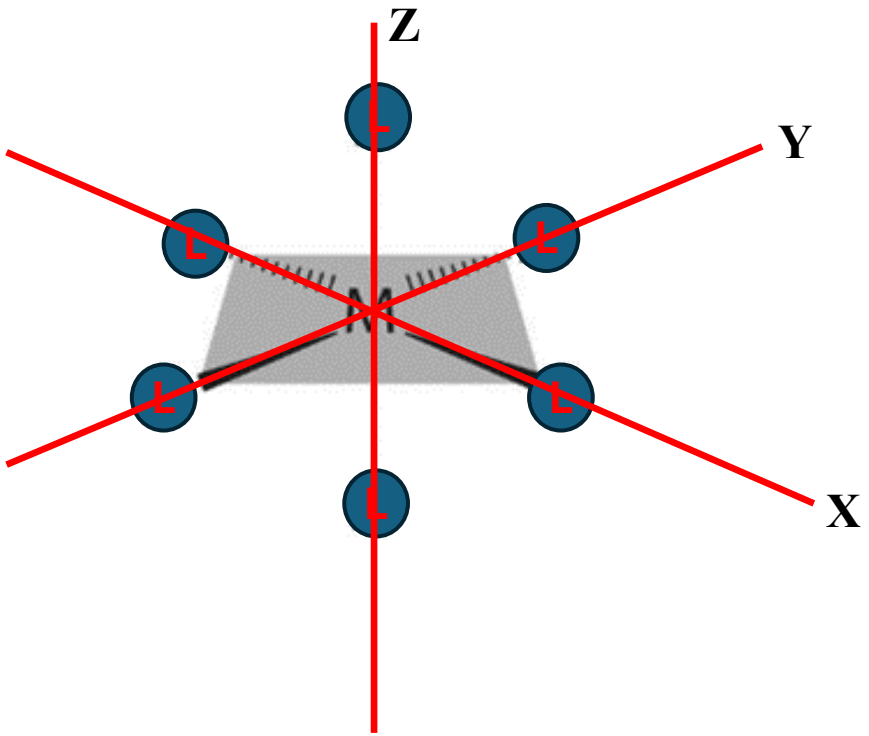
Isolated Metal Atom

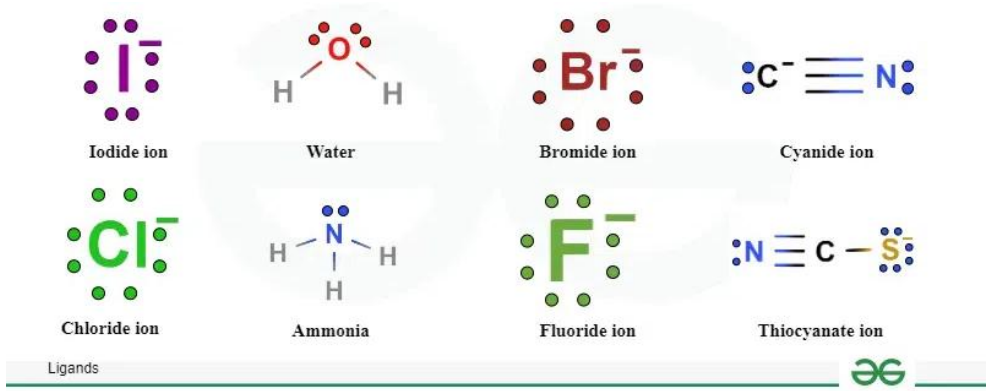


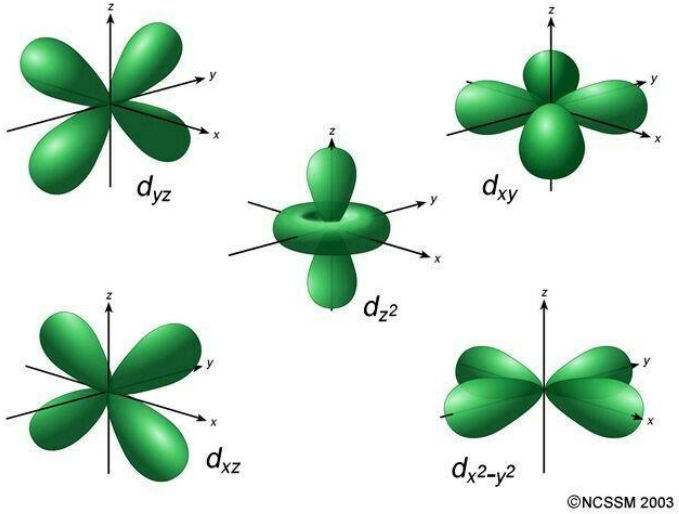
Degeneracy splits

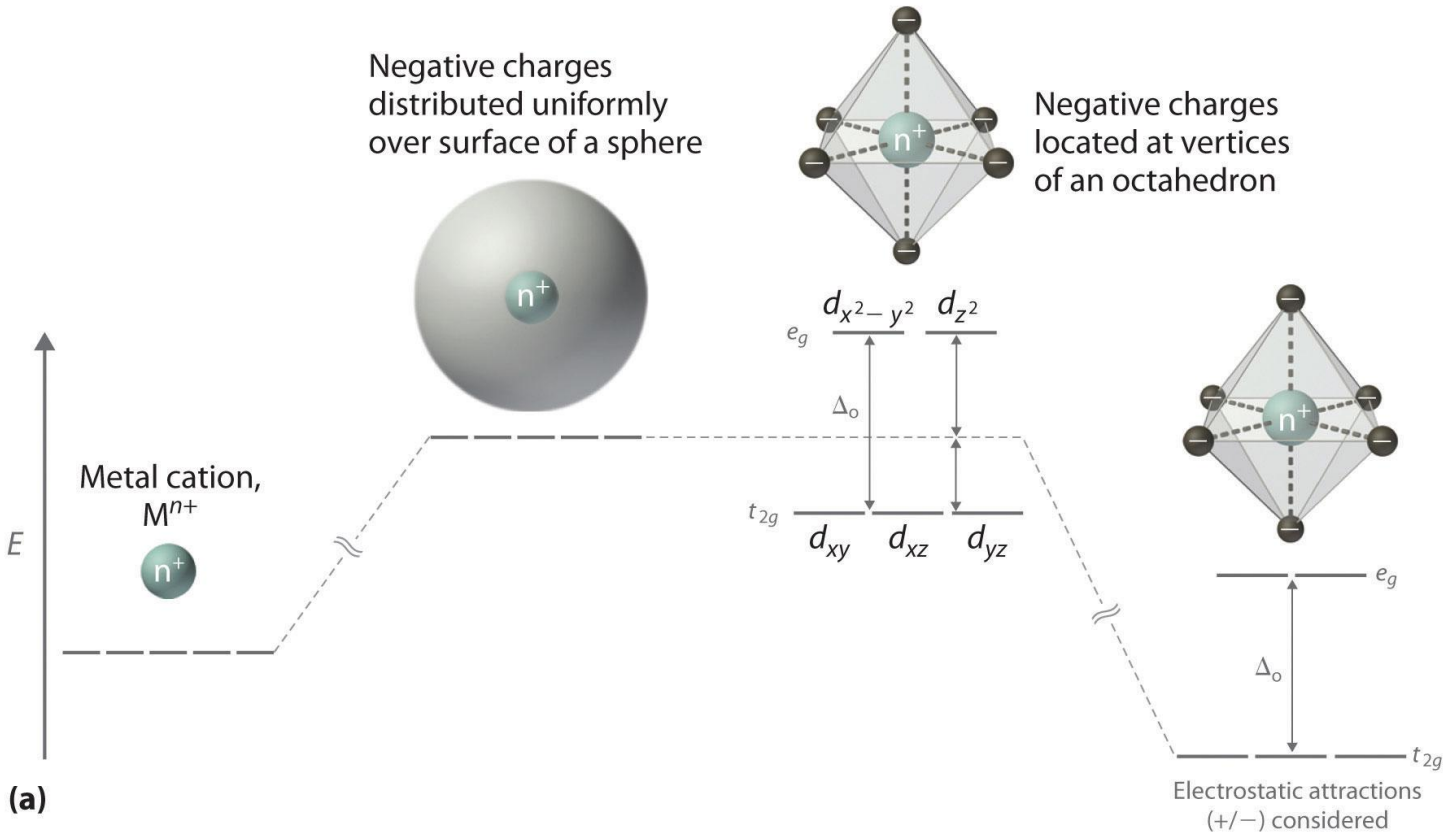
Crystal Field Splitting

Crystal field splitting in octahedral coordination complexes

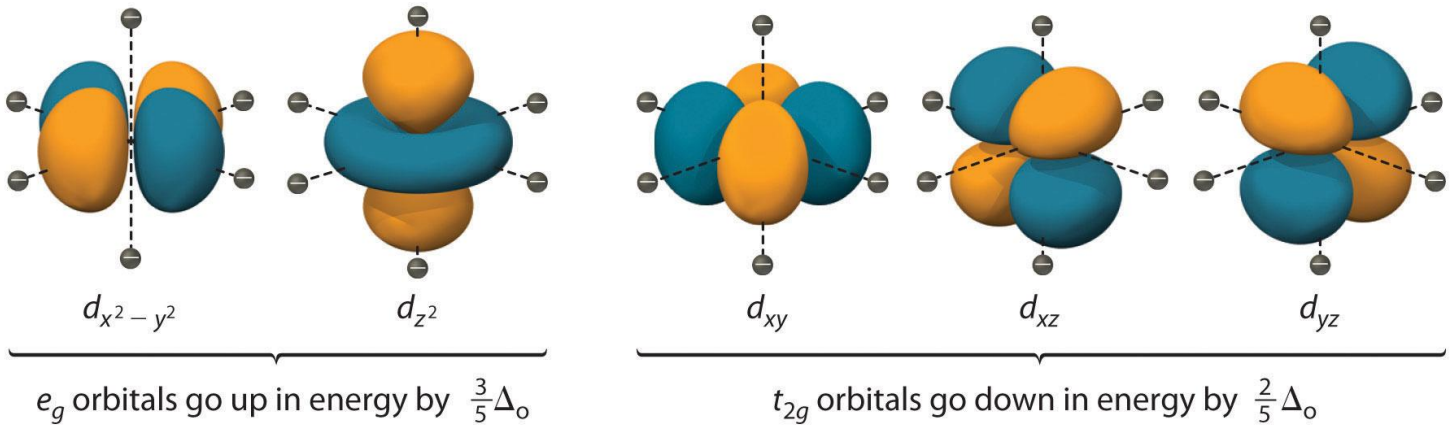




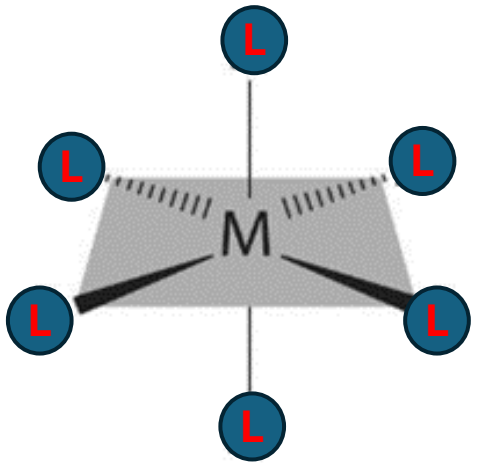


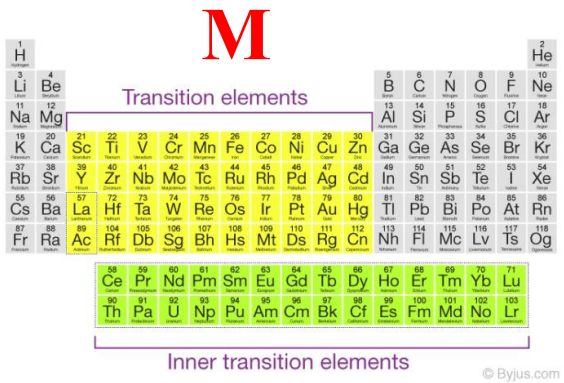

Negative charges distributed uniformly over surface of a sphere
Negative charges located at vertices of an octahedron
Metal cation, M^{n+}
E
e_g
$d_{x^2-y^2}$
d_{z^2}
Δ_o
t_{2g}
d_{xy}
d_{xz}
d_{yz}
e_g
Δ_o
t_{2g}
Electrostatic attractions (+/−) considered

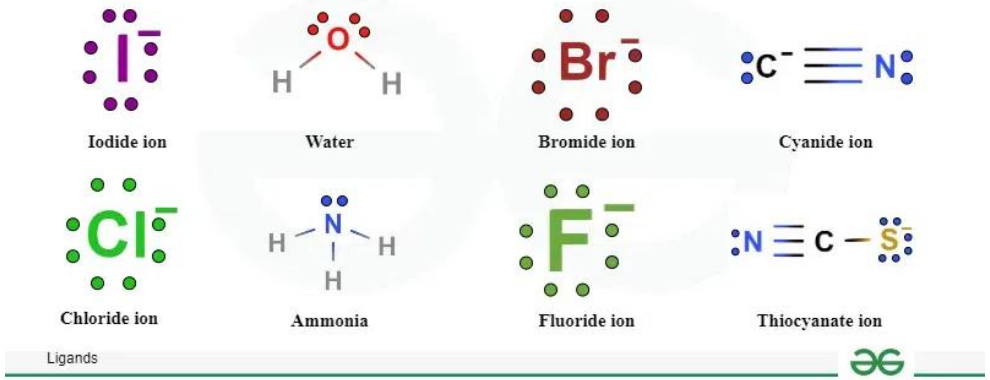
(a)

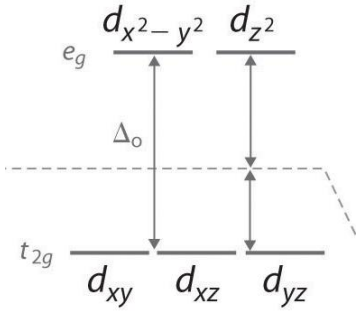

$d_{x^2-y^2}$
d_{z^2}
d_{xy}
d_{xz}
d_{yz}
e_g orbitals go up in energy by $\frac{3}{5}\Delta_o$
t_{2g} orbitals go down in energy by $\frac{2}{5}\Delta_o$

Spectrochemical series









Ryutaro Tsuchida

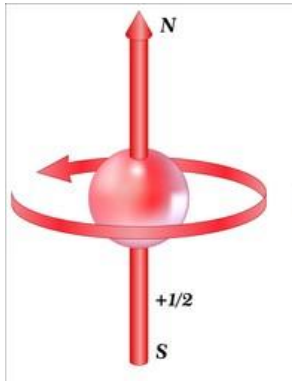
1930

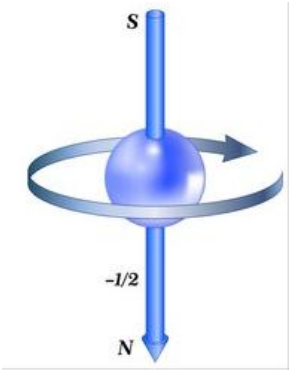
Octahedral cobalt(III) complexes

Spectrochemical series

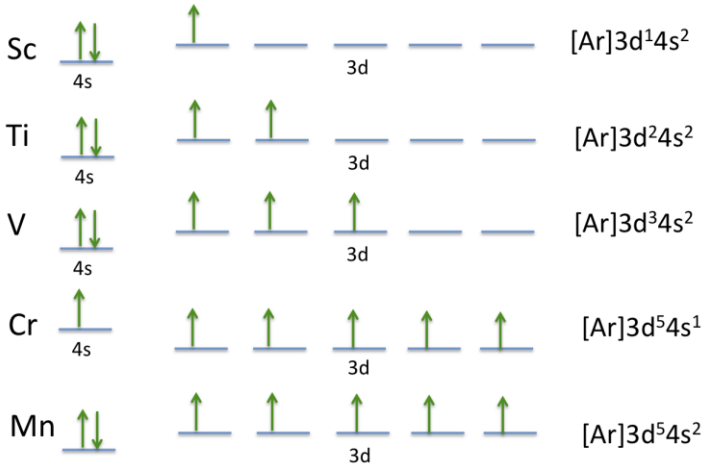
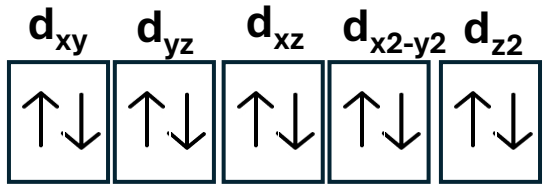
$I^- < Br^- < S^{2-} < \mathbf{S}CN^- < Cl^- < O{=}N{-}\mathbf{O}^- < N_3^- < F^- <$
$OH^- < C_2O_4^{2-} < O^{2-} < H_2O < \mathbf{N}CS^- < CH_3C{\equiv}N <$
$EDTA^{4-} < py < NH_3 < en < bipy < \mathbf{N}O_2^- < PPh_3 < \mathbf{C}N^- < CO$

(The donor atoms are indicated by bold letters.)





SPIN ANGULAR MOMENTUM



Pairing energy